The documentation and process conversion measures necessary to comply with this revision shall be completed by 13 July 2021.

INCH-POUND

MIL-PRF-19500/556L
w/AMENDMENT 2
13 April 2021
SUPERSEDING
MIL-PRF-19500/556L
w/AMENDMENT 1
7 June 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT, N-CHANNEL, SILICON,
TYPES 2N6782, 2N6784, AND 2N6786,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-channel, enhancement-mode, MOSFET, power transistor. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device.

1.2 Package outlines. The device package outlines are as follows: TO-205AF in accordance with figure 1 and leadless chip carrier (LCC) in accordance with figure 2 for all encapsulated device types. See figures 3 and 4 for all unencapsulated devices.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T (1) T_C = +25°C	P_T T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} (2) (3) T_C = +25°C	I_{D2} (2) T_C = +100°C	I_S	I_{DM} (4)	T_J and T_{STG}	V_{ISO} 70,000 foot altitude
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)	°C	V dc
2N6782, U	15	0.8	100	100	±20	3.5	2.25	3.50	14.0	-55 to	
2N6784, U	15	0.8	200	200	±20	2.25	1.50	2.25	9.0	+150	
2N6786, U	15	0.8	400	400	±20	1.25	0.80	1.25	5.5		400

(1) Derate linearly 0.12 W/°C for T_C > +25°C.
(2) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(3) See figure 5, maximum drain current graph.
(4) I_{DM} = 4 x I_{D1} as calculated in note 2.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(TH)}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(ON)}$ (1) V_{GS} = 10 V dc		$R_{\theta JC}$ max (2)
					T_J = +25°C at I_{D2}	T_J = +150°C at I_{D2}	
	V dc	V dc		μA dc	ohm	ohm	°C/W
		Min	Max				
2N6782, U	100	2.0	4.0	25	0.60	1.20	8.33
2N6784, U	200	2.0	4.0	25	1.50	3.15	8.33
2N6786, U	400	2.0	4.0	25	3.60	9.00	8.33

(1) Pulsed (see 4.5.1).
(2) See figure 6, thermal impedance curves.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

1.5.2 JAN certification mark and quality level for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

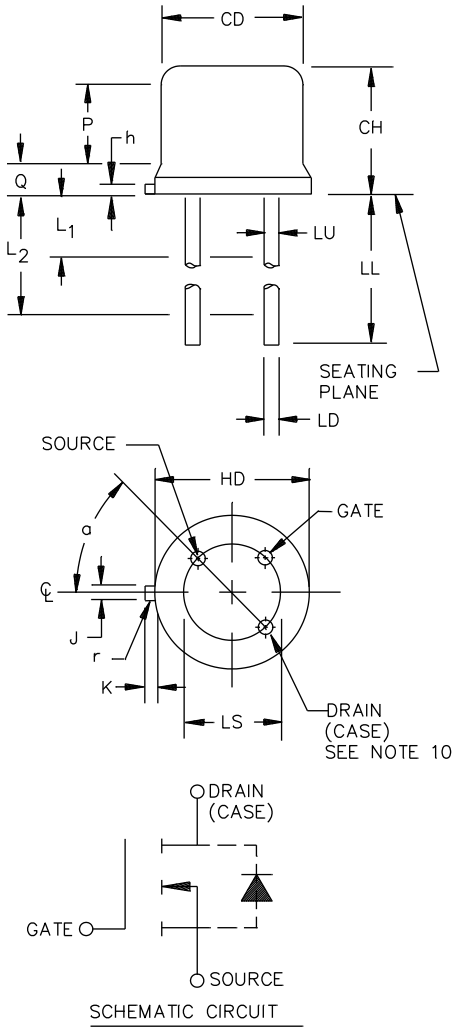
1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "6782", "6784", and "6786".

1.5.3.3 Suffix letters. No suffix letters are used on devices that are packaged in the TO-205AF package of figure 1. The suffix letter "U" is used on devices that are packaged in the LCC package of figure 2.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A" and "B" (see figures 3 and 4).

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SCHEMATIC CIRCUIT

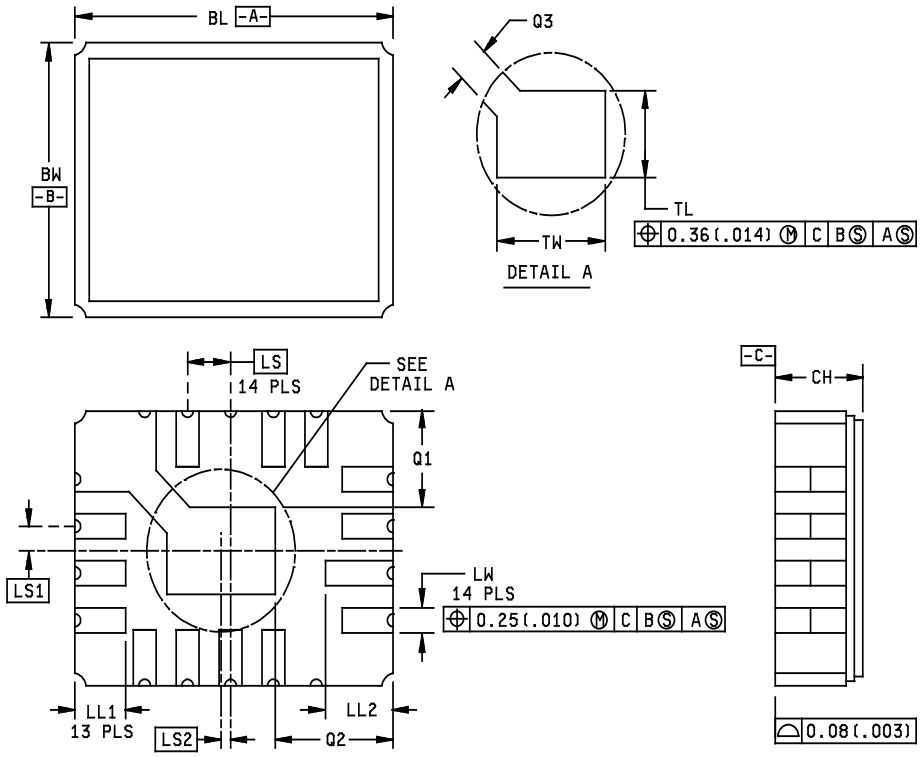
Symbol	Dimensions				
	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CD	.315	355	8.01	9.01	
CH	.160	.180	4.06	4.57	
HD	.340	.370	8.64	9.40	
h	.009	.041	0.23	1.04	
J	.028	.034	0.71	0.86	2
k	.029	.045	0.74	1.14	3
LD	.016	.019	0.41	0.48	7, 8
LL	.500	.750	12.7	19.05	7, 8
LS	.200 TP		5.08 TP		6
LU	.016	.019	0.41	0.48	7, 8
L1		.050		1.27	7, 8
L2	.250		6.35		7, 8
P	.100		2.54		5
Q		.050		1.27	4
r		.010		0.25	9
α	45° TP		45° TP		6

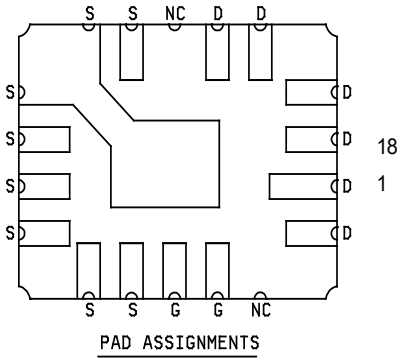
NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Beyond radius (r) maximum, J shall be held for a minimum length of .011 (0.28 mm).
3. Dimension k measured from maximum HD.
4. Outline in this zone is not controlled.
5. Dimension CD shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
* 7. LU applies between L_1 and L_2. LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. Radius (r) applies to both inside corners of tab.
10. Drain is electrically connected to the case.
* 11. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for TO-205AF.

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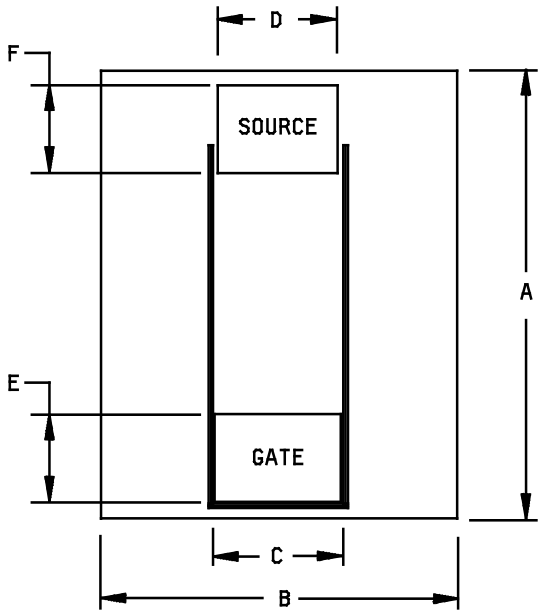
Dimensions				
Ltr.	Inches		Millimeters	
	Min	Max	Min	Max
BL	.345	.360	8.76	9.14
BW	.280	.295	7.11	7.49
CH	.095	.115	2.41	2.92
LL1	.040	.055	1.02	1.40
LL2	.055	.065	1.40	1.65
LS	.050 BSC		1.27 BSC	
LS1	.025 BSC		0.635 BSC	
LS2	.008 BSC		0.203 BSC	
LW	.020	.030	0.51	0.76
Q1	.105 REF		2.67 REF	
Q2	.120 REF		3.05 REF	
Q3	.045	.055	1.14	1.40
TL	.070	.080	1.78	2.03
TW	.120	.130	3.05	3.30

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
* 3. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.
4. Ceramic package only.

FIGURE 2. Physical dimensions for LCC.

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2N6782, 2N6784, and 2N6786

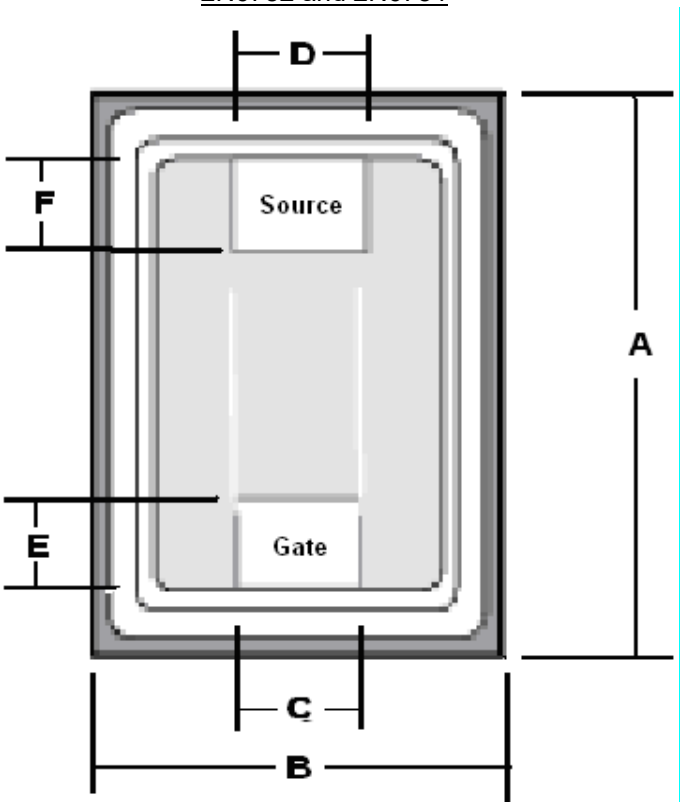


Ltr	Dimensions - 2N6782				Dimensions - 2N6784				Dimensions - 2N6786			
	Inches		Millimeters		Inches		Millimeters		Inches		Millimeters	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
A	0.0759	0.0917	1.929	2.329	0.082	0.092	2.08	2.34	0.101	0.111	2.55	2.81
B	0.0528	0.0685	1.340	1.740	0.062	0.072	1.57	1.83	0.071	0.081	1.81	2.07
C	0.0258	0.0262	0.655	0.665	0.020	0.030	0.50	0.76	0.020	0.030	0.50	0.76
D	0.0245	0.0249	0.623	0.633	0.019	0.029	0.47	0.73	0.019	0.029	0.47	0.73
E	0.0174	0.0178	0.442	0.542	0.012	0.022	0.31	0.57	0.012	0.022	0.31	0.57
F	0.0185	0.0189	0.469	0.479	0.013	0.023	0.32	0.58	0.013	0.023	0.32	0.58

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and the back contact is the drain. The top metal is aluminum.
5. Die thickness is .0187 inch (0.475 mm) ±.0050 inch (0.130 mm).

* FIGURE 3. JANHCA and JANKCA die dimensions.

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Ltr	Dimensions - 2N6782 and 2N6784			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.082	.089	2.08	2.26
B	.062	.066	1.58	1.68
C	.019	.021	0.48	0.53
D	.022	.024	0.56	0.61
E	.012	.014	0.30	0.36
F	.013	.015	0.33	0.38

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and the back contact is the drain. The top metal is aluminum.
5. Die thickness is .015 inch (0.38 mm) ±.001 inch (0.025 mm).

FIGURE 4. JANHCB and JANKCB die dimensions (2N6782, 2N6784).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.
* MIL-STD-883 - Test Method Standard for Microcircuits.

* (Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

nC - nano coulomb.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 (TO-205), 2 (LCC), and 3 and 4 (die) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

* 3.4.3 Silicone die coating. The use of a silicone die coat requires a successful completion of MIL-STD-883, method 5011 on each silicone lot for its intended applications, and as part of the full MIL-PRF-19500 qualification process.

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3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-PRF-19500.

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4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500) (1) (2)	Measurement	
	JANS level	JANTX and JANTXV levels
(3)	Gate stress test (see 4.3.1).	Gate stress test (see 4.3.1).
(3) (4)	Unclamped inductive switching, method 3470 of MIL-STD-750 (see 4.3.2), optional.	Unclamped inductive switching, method 3470 of MIL-STD-750 (see 4.3.2), optional.
(3) 3c	Method 3161 of MIL-STD-750 (see 4.3.3).	Method 3161 of MIL-STD-750 (see 4.3.3).
* 5	Method 2052 of MIL-STD-750, PIND (see MIL-PRF-19500 and 4.3.5)	Not applicable
9	Subgroup 2 of table I herein.	Not applicable.
10	Method 1042 of MIL-STD-750, test condition B.	Method 1042 of MIL-STD-750, test condition B.
11	Subgroup 2 of table I herein; ΔI_{GSSF1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 25 µA dc or ±100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein.
12	Method 1042 of MIL-STD-750, test condition A, t = 240 hours.	Method 1042 of MIL-STD-750, test condition A.
13	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value. $\Delta V_{GS(TH)1}$= ± 20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GSSF1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value. $\Delta V_{GS(TH)1}$= ± 20 percent of initial value.

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.
(4) This test is optional in screening if performed in table I, subgroup 5.

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4.3.1 Gate stress test. Apply V_{GS} = ±30 V minimum for t = 250 µs minimum.

4.3.2 Unclamped inductive switching.

a. Peak current (I_D)..... rated I_{D1}.

b. Peak gate voltage (V_{GS}) ..... 10 V.

c. Gate to source resistor (R_{GS})..... 25 Ω ≤ R_{GS} ≤ 200 Ω.

d. Initial case temperature (T_C)..... +25°C +10°C, -5°C.

e. Inductance (L) ..... 100 µH minimum.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Pulse repetition rate ..... None.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (and V_H where appropriate). See table II, group E, subgroup 4 herein.

4.3.4 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100-percent probed in accordance with table I, subgroup 2, except test current shall not exceed 20 A.

* 4.3.5 PIND. Not applicable in screening when devices are processed using alternative method and flow requirements approved by the qualifying activity, that includes incorporating the use of certified clean processing and silicone die coat. Instead, the PIND test performance shall be performed in group B3 and group C3, on a lot sample basis. PIND failures detected in group B or C will represent lot jeopardy and shall be evaluated for root cause and lot integrity.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for quality conformance inspection in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein.

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4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

	Subgroup	Method	Condition
	B3	1051	Test condition G.
*	B3	2052	PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	B4	1042	Intermittent operation life, test condition D. The heating cycle shall be 1 minute minimum.
*	B5	1042	Accelerated steady-state operation life; test condition A, V_{DS} = rated T_A = +175°C, t = 120 hours minimum; or, T_A = +150°C, t = 240 hours minimum. Read and record $V_{(BR)DSS}$ (pre and post) at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10 percent and I_{DSS} shall not exceed 25 μA.
*			Accelerated steady-state gate bias; condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours minimum; or, T_A = +150°C, t = 48 hours minimum.
	B5	2037	Bond strength; test condition D.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition G.
B3	1042	Intermittent operation life, test condition D. The heating cycle shall be 1 minute minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

	Subgroup	Method	Condition
	C2	2036	Test condition E (Not required for LCC).
*	C3	2052	PIND, required if not performed in screening. (JANS only, 22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	C5	3161	See 4.5.2, $R_{\theta JC(max)}$ = 8.33°C/W.
	C6	1042	Intermittent operation life, test condition D, 6,000 cycles. The heating cycle shall be 1 minute minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

* 4.5.2 Thermal resistance. The thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW} (and V_H where appropriate). See table II, group E, subgroup 4 herein.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.3	$Z_{\theta JC}$			°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = 1.0 mA dc, bias condition C	$V_{(BR)DSS}$			
2N6782, 2N6782U				100		V dc
2N6784, 2N6784U				200		V dc
2N6786, 2N6786U				400		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA dc	$V_{GS(TH)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 V dc, bias condition C, V_{DS} = 0	I_{GSSF1}		+100	nA dc
Gate current	3411	V_{GS} = -20 V dc, bias condition C, V_{DS} = 0	I_{GSSR1}		-100	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, condition A, pulsed (see 4.5.1), I_D = I_{D2}	$r_{DS(on)1}$			
2N6782, 2N6782U					0.60	ohm
2N6784, 2N6784U					1.50	ohm
2N6786, 2N6786U					3.60	ohm
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, condition A, pulsed (see 4.5.1), I_D = I_{D1}	$r_{DS(on)2}$			
2N6782, 2N6782U					0.61	ohm
2N6784, 2N6784U					1.60	ohm
2N6786, 2N6786U					3.70	ohm
Forward voltage (source drain diode)	4011	Condition A, I_D = I_{D1}, V_{GS} = 0 V dc	V_{SD}			
2N6782, 2N6782U					1.5	V
2N6784, 2N6784U					1.5	V
2N6786, 2N6786U					1.4	V

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		$T_C = T_J = +125°C$				
Gate current	3411	V_{GS} = +20 V dc and -20 V dc, bias condition C, V_{DS} = 0	I_{GSS2}		± 200	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		0.25	mA dc
Gate to source voltage (thresholds)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA dc	$V_{GS(TH)2}$	1.0		V dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1), $I_D = I_{D2}$, condition A	$r_{DS(on)3}$			
2N6782, 2N6782U					1.08	ohm
2N6784, 2N6784U					2.81	ohm
2N6786, 2N6786U					7.92	ohm
Low-temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA dc	$V_{GS(TH)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	$I_D = I_{D1}$, V_{GS} = 10 V dc, R_G = 7.5Ω, V_{DD} = 50 percent of rated V_{DS}				
Turn-on delay time			$t_{d(on)}$			
2N6782, 2N6782U					15	ns
2N6784, 2N6784U					15	ns
2N6786, 2N6786U					15	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Rise time			t_r			
2N6782, 2N6782U					25	ns
2N6784, 2N6784U					20	ns
2N6786, 2N6786U					20	ns
Turn-off delay time			$t_{d(off)}$			
2N6782, 2N6782U					25	ns
2N6784, 2N6784U					30	ns
2N6786, 2N6786U					35	ns
Fall time			t_f			
2N6782, 2N6782U					20	ns
2N6784, 2N6784U					20	ns
2N6786, 2N6786U					30	ns
Subgroup 5						
Single pulse unclamped Inductive switching 3/	3470	See 4.3.2				
Safe operating area test	3474	See figure 7; t_p = 10 ms minimum, V_{DS} = 80 percent of maximum rated V_{DS}, ($V_{DS} \leq$ 200)				
Electrical measurements		See table I, subgroup 2 herein.				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
Test 1						
On-state gate charge			$Q_{g(on)}$			
2N6782, 2N6782U					8.1	nC
2N6784, 2N6784U					8.6	nC
2N6786, 2N6786U					12	nC

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued						
Test 2						
Gate to source charge			Q_{gs}			
2N6782, 2N6782U					1.7	nC
2N6784, 2N6784U					1.5	nC
2N6786, 2N6786U					1.8	nC
Test 3			Q_{gd}			
Gate to drain charge						
2N6782, 2N6782U					4.5	nC
2N6784, 2N6784U					5.5	nC
2N6786, 2N6786U					7.6	nC
Reverse recovery time	3473	$d_i/ d_t \leq 100$ A/μs, condition A, $V_{DD} \leq 50$ V, $I_D = I_{D1}$	t_{rr}			
2N6782, 2N6782U		$I_F = 3.5$ A			180	ns
2N6784, 2N6784U		$I_F = 2.25$ A			350	ns
2N6786, 2N6786U		$I_F = 1.25$ A			540	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.
3/ This test is optional if performed as a 100 percent screen.

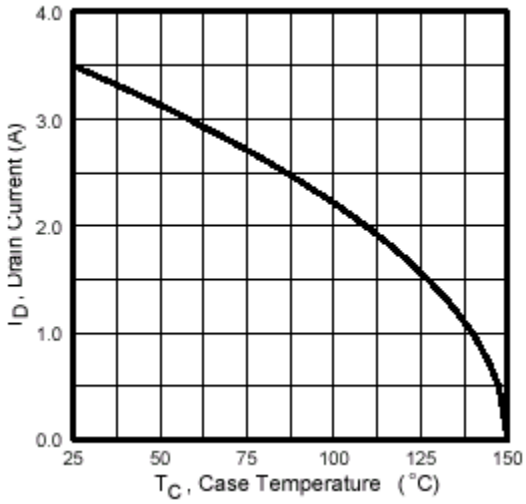
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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

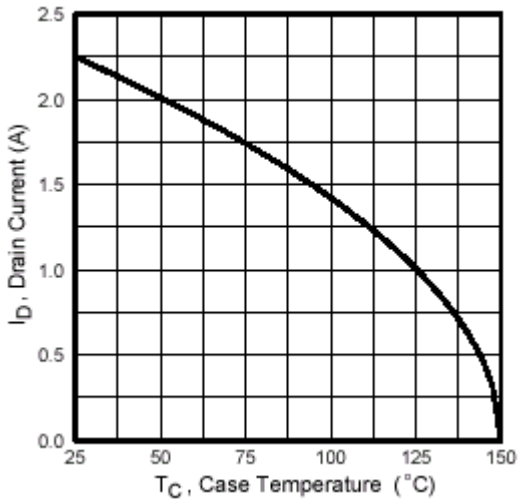
Inspection 1/	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	Test condition G, 500 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2	
Subgroup 2 2/			45 devices c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state gate bias	1042	Condition B, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			3 devices c = 0
Barometric pressure	1001	Condition C, 2N6786 and 2N6786U only	
Subgroup 10			22 devices c = 0
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer.	

1/ JANHC and JANKC devices are qualified in accordance with MIL-PRF-19500.
2/ A separate sample may be pulled for each test.

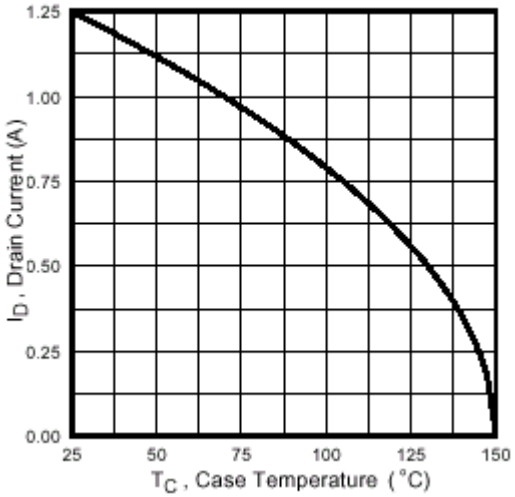
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2N6782, 2N6782U



2N6784, 2N6784U



2N6786, 2N6786U

FIGURE 5. Maximum drain current versus case temperature graphs.

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2N6782, 2N6782U, 2N6784, 2N6784U, 2N6786, 2N6786U

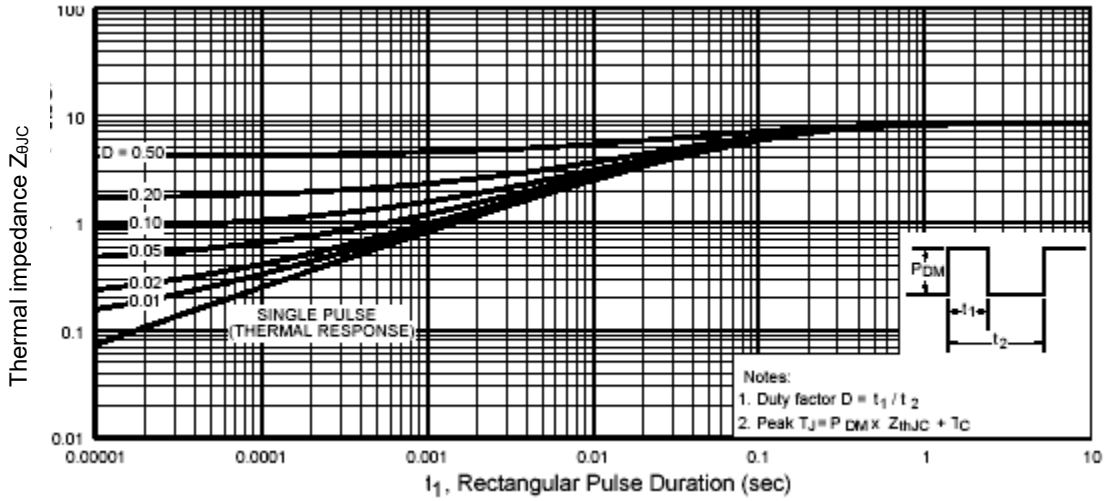
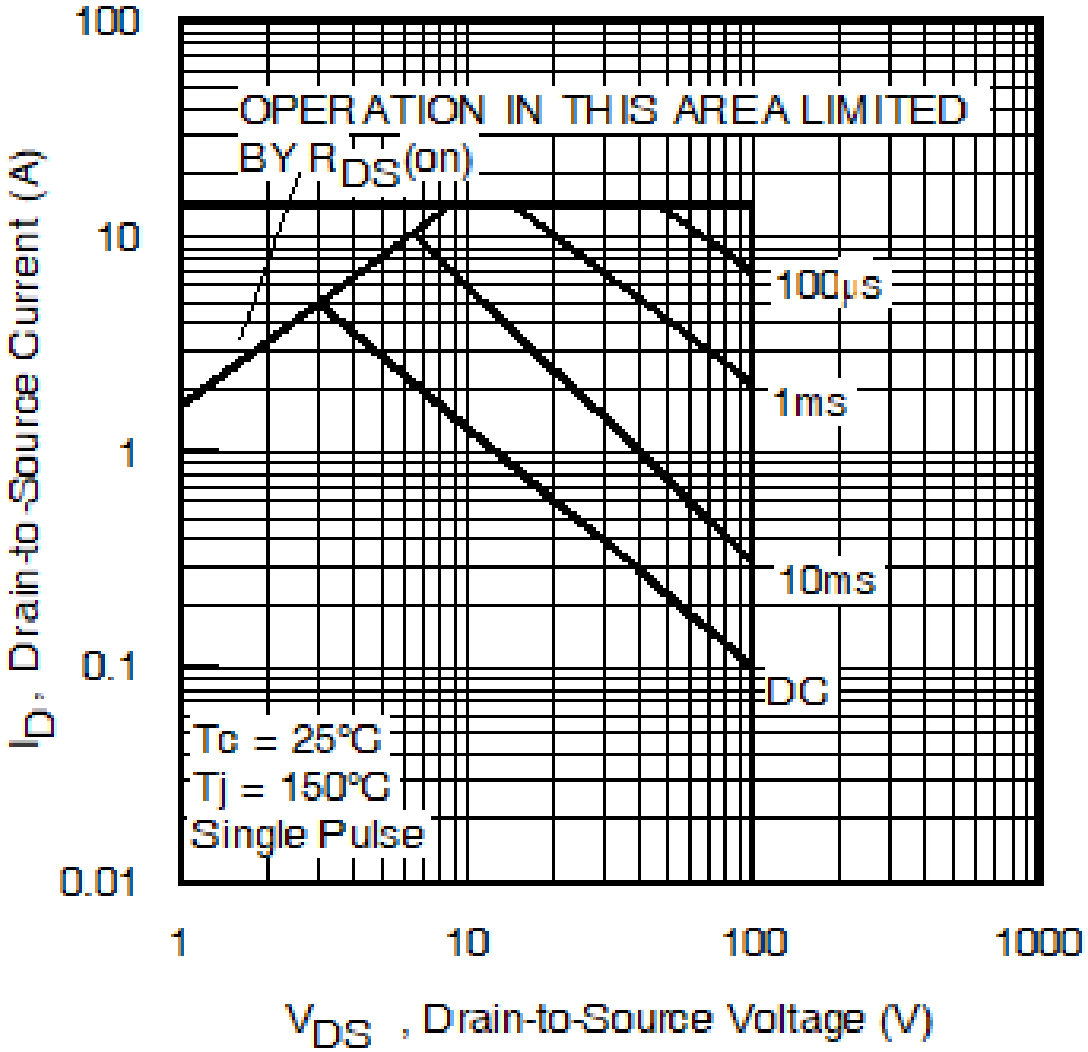


FIGURE 6. Thermal impedance curve.

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* FIGURE 7. Maximum safe operating area.

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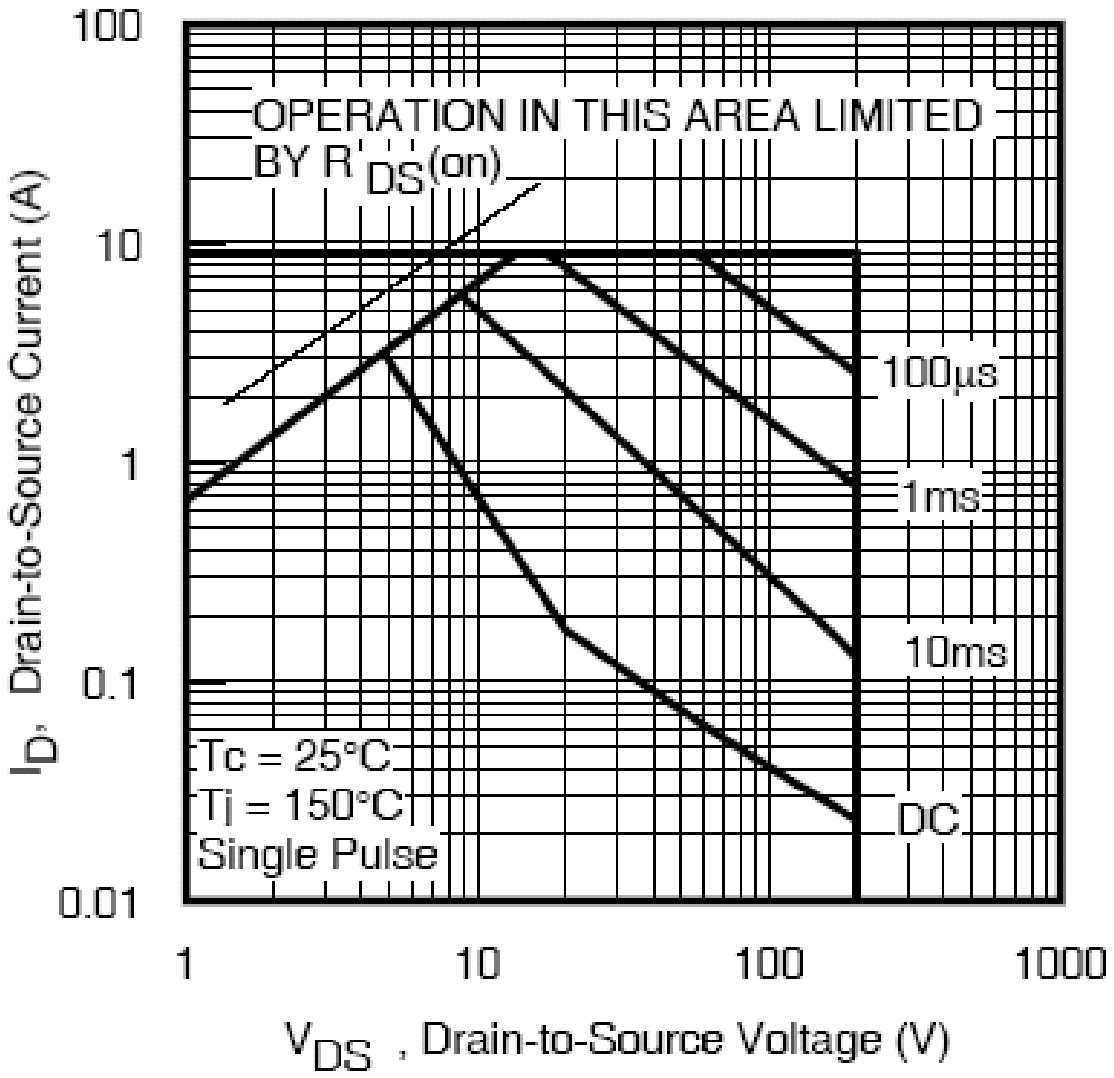


FIGURE 7. Maximum safe operating area - Continued.

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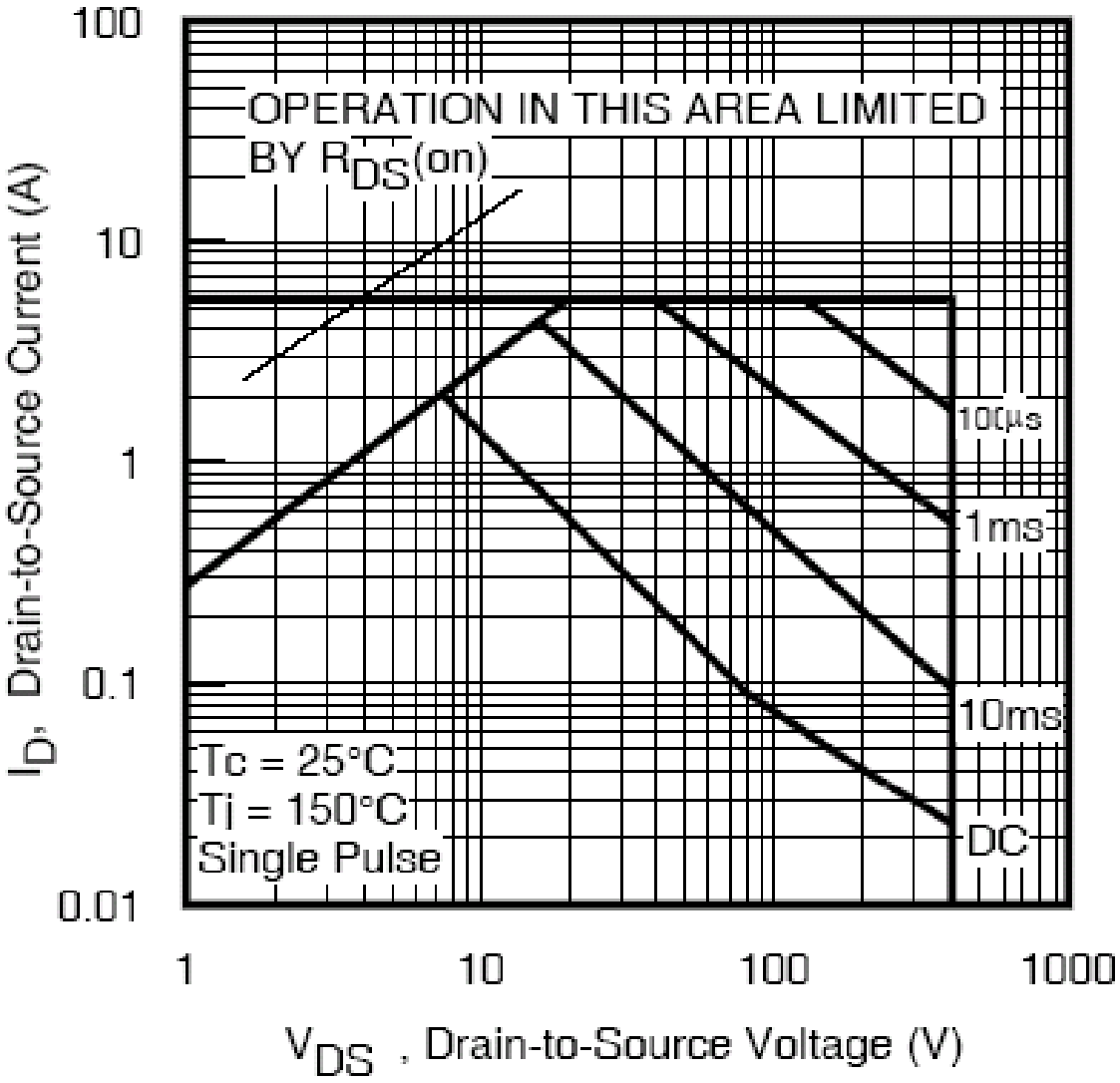


FIGURE 7. Maximum safe operating area - Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.6.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 3 and 4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

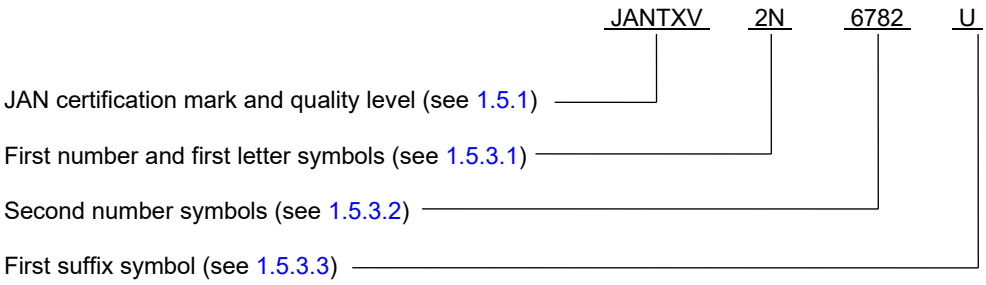
6.4 Cross-reference and complement list. Parts from this specification may be used to replace the following commercial Part or Identifying Number (PIN). The term PIN is equivalent to the term part number which was previously used in this specification.

Preferred types	Commercial types
2N6782	IRFF110, IRFF111, IRFF112, IRFF113
2N6784	IRFF210, IRFF211, IRFF212, IRFF213
2N6786	IRFF310, IRFF311, IRFF312, IRFF313
2N6782U	IRFE110, IRFE111, IRFE112, IRFE113
2N6784U	IRFE210, IRFE211, IRFE212, IRFE213
2N6786U	IRFE310, IRFE311, IRFE312, IRFE313

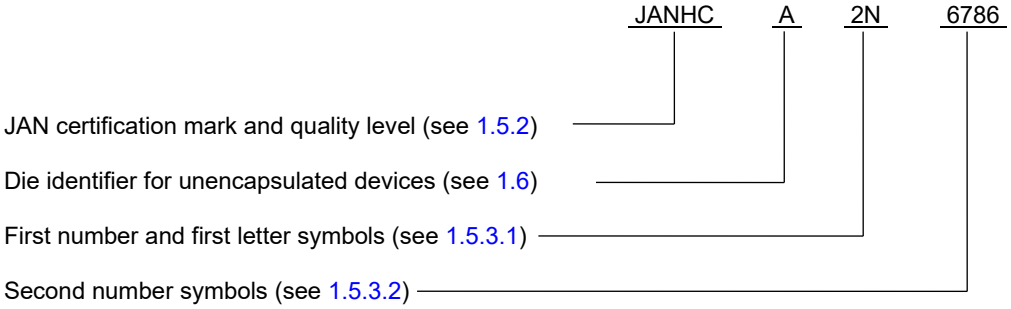
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6.5 PIN construction example.

6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



6.5.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.6 List of PINs.

6.6.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N6782	JANTX2N6782	JANTXV2N6782	JANS2N6782
JAN2N6782U	JANTX2N6782U	JANTXV2N6782U	JANS2N6782U
JAN2N6784	JANTX2N6784	JANTXV2N6784	JANS2N6784
JAN2N6784U	JANTX2N6784U	JANTXV2N6784U	JANS2N6784U
JAN2N6786	JANTX2N6786	JANTXV2N6786	JANS2N6786
JAN2N6786U	JANTX2N6786U	JANTXV2N6786U	JANS2N6786U

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6.6.2 List of PINs for unencapsulated devices. The following is a list of possible PINs for unencapsulated devices available on this specification sheet. The qualified die suppliers with the applicable letter version (example, JANHCA2N6786) will be identified on the QML.

JANC ordering information		
PIN	Manufacturer	
	69210	43611
2N6782	JANHCA2N6782 JANKCA2N6782	JANHCB2N6782 JANKCB2N6782
2N6784	JANHCA2N6784 JANKCA2N6784	JANHCB2N6784 JANKCB2N6784
2N6786	JANHCA2N6786 JANKCA2N6786	

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
* NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
* Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2021-009)

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